MIL-S-22215A(AS)
17 May 1968
SUPERSEDING
MIL-S-22215(Aer)
2 October 1959

MILITARY SPECIFICATION

SILVER-COPPER-CADMIUM-NICKEL ALLOY

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers the requirements for a silver-base alloy containing copper, cadmium and nickel, to be used in electrical contacts and in structures where strength, hardness, and high thermal conductivity are required (see 6.1).

1.2 Classification - Material shall be of one grade (see Table I): and shall be furnished in the following classes and conditions, as specified (see 6.2).

Classes	Conditions	
I - Sheet and strip	1- Annealed	5- Hard
II - Wire and rod	2- 1/8 Hard	6- Spring
	3—1/4 Hard	7- Extra spring
	4- 1/2 Hard	

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-B-121	Barrier Material, Grease-proofed, Waterproofed, Flexible

FSC 9535

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STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publications - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Circular No. 31	International Annealed Copper Standard

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

American Society for Testing and Materials

B-193	Resistivity of Electrical Conductor Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Chemical composition - The material shall conform to the chemical requirements shown in Table I.

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TABLE I

CHEMICAL COMPOSITION

Silver	86.8 ± 1.0%
Copper	7.5 ± 0.5%
Cadmium	5.5 ± 0.5%
Nickel	0.20 ± 0.05%
Impurities	
Iron	0.05% max.
Zinc	0.06% max.
Lead	0.03% max.
Other impurities	0.11% max.
Total impurities	0.15% max.

3.2 Physical properties - The alloy shall be capable of meeting the requirements of Table II.

TABLE II

MECHANICAL PROPERTIES

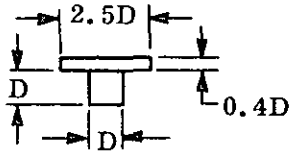
Condition	Reduction in B&S Gauge Nos	Hardness Rockwell 30t		Tensile Strength p.s.i.		Elongation % in 2" Min.
		Min.	Max.	Min.	Max.	
(Sheets and Strip)						
1	0	30	44	37,000	47,000	28
4	2	57	65	51,000	60,000	5
5	4	64	70	61,000	68,000	3
6	8	67	72	70,000	77,000	1
7	10	69	74	73,000	80,000	
(Wire and Rod)						
1	0			37,000	47,000	28
2	1/2			44,000	54,000	10
3	1			51,000	61,000	5
5	4			71,000	78,000	2
6	8			78,000	85,000	1

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3.3 Bend requirements - All sheet and strip shall be capable of withstanding a 180 degree bend around a radius equal to the thickness of the strip without any signs of flaking or fracture. This requirement shall apply only to bends which are at 90 degrees to the direction of rolling.

3.4 Twist requirements - All wire and rod, having temper of 1/4 hard or less, shall be capable of withstanding a twist of 7 forward and 7 reverse 360 degree turns within a length of 3 inches without exhibiting seams, slivers, or other discontinuities that would be detrimental.

3.5 Headability - All wire annealed to the temper of 1/4 hard or less shall be capable of being cold headed into the following shape:



3.6 Alloying - The microconstituents of the alloy shall be uniformly distributed so that segregation is not present in the alloy.

3.7 Dimensions - The width, length, thickness, and diameter of the material shall be as specified in the individual contract (see 6.2).

3.7.1 Dimensional tolerances, sheet and strip - Unless otherwise specified, dimensional tolerances shall conform to the requirements shown in Table III.

TABLE III

DIMENSIONAL TOLERANCES FOR SHEET AND STRIP 1/

Thickness Tolerances			
Thickness Above Min. Inclusive Max.	Width		
	up to 9	over 9 to 14	over 14 to 20 inclusive
Up to 0.004	±0.0003	±0.0005	
0.004 to 0.006	±0.0004	±0.0006	
0.006 to 0.009	±0.0006	±0.0008	±0.0012
0.009 to 0.013	±0.0008	±0.0010	±0.0015
0.013 to 0.017	±0.0010	±0.0014	±0.0018
0.017 to 0.021	±0.0012	±0.0016	±0.0020

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TABLE III (Cont'd)

Thickness Tolerances

Thickness Above Min. Inclusive Max.	Width		
	up to 9	over 9 to 14	over 14 to 20 inclusive
0.021 to 0.026	±0.0015	±0.0020	±0.0025
0.026 to 0.037	±0.0020	±0.0020	±0.0025
0.037 to 0.050	±0.0020	±0.0020	±0.0025
0.050 to 0.073	±0.0025	±0.0025	±0.0030
0.073 to 0.130	±0.0025	±0.0025	±0.0030
0.130 to 0.187	±0.0030	±0.0030	±0.0030
0.187 to 0.205	±0.0030	±0.0030	±0.0035
0.205 to 0.300	±0.0030	±0.0030	±0.0035

Width Tolerances

Width	Slit Metal			Square Sheared	Sawed	
	Up to 0.062	Over 0.062 to 0.125	Over 0.125		Up to 0.5	Over 0.5
Up to 2	±0.003	±0.008	±0.015	±0.062	±0.031	±0.062
2 to 6	±0.003	±0.015	±0.015	±0.062	±0.031	±0.062
6 to 20	±0.005	±0.015	±0.015	±0.062	±0.031	±0.062

Straightness Tolerance - Maximum edgewise curvature in any 6 feet

Width	Slit Metal			Square Sheared		Sawed
	Coiled	Flat	Flat and Straightened	Up to 0.125	0.125 to 0.250	
Up to 2 inclusive	0.500	0.500	0.250	0.025	0.031	0.031
Over 2 to 6 inclusive	0.500	0.500		0.025	0.031	0.031
Over 6 to 13 inclusive	0.500	0.500		0.031	0.062	0.050

1/ All dimensions in inches.

3.7.2 Dimensional tolerances, wire and rod - Dimensional tolerances for wire and rod shall conform to the requirements shown in Table IV.

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TABLE IV

DIMENSIONAL TOLERANCES FOR WIRE AND ROD 1/

Diameter	Tolerances
0.012 to 0.030	±0.0002
0.030 to 0.060	±0.0003
0.060 to 0.187	±0.0005
0.187 to 0.250	±0.0010
0.250 to 0.375	±0.0015
0.375 to 0.750	±0.0020

1/ All dimensions in inches

3.8 Resistivity and electrical conductivity - The resistivity and electrical conductivity of the alloy shall conform to the requirements outlined in Table V.

TABLE V

RESISTIVITY AND ELECTRICAL CONDUCTIVITY

Condition	ρ, Resistivity Microhms/cm^3	Electrical Conductivity % I.A.C.S. 1/
	Max.	Min.
	Sheet & Strip	
1	4.310	40
4	4.925	35
5	4.925	35
6	4.925	35
7	4.925	35
	Wire & Rod	
1	4.310	40
2	4.310	40
3	4.925	35
5	4.925	35
6	4.925	35

1/ International Annealed Copper Standard - Bureau of Standards Circular No. 31.

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3.9 Surface finish - The alloy shall be free of dull surfaces, die marks and seams.

3.10 Identification marking - The alloy shall be legibly marked with the manufacturer's name, brand or trademark, the condition and the specification number. The marking shall be such as not to adversely affect use of the material.

3.11 Workmanship - The alloy shall be free of any imperfections which will cause any inadequacy in its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling -

4.2.1 Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier for acceptance shall be in accordance with the provisions set forth in MIL-STD-105.

4.3 Lot size - As applied to inspection of units of product, the term "lot" shall mean "inspection lot", i.e., a collection of units of product submitted by a supplier for acceptance. Unless otherwise specified, the number of units of product in "inspection lots" may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.4 Classification of tests - The inspection and testing of the silver alloy shall be classified as follows:

4.4.1 Quality conformance tests - Quality conformance tests are those performed on production lots by the manufacturer.

4.4.1.1 Contractor's tests - The contractor shall conduct quality conformance tests as specified in 4.4.1 to assure that each lot of silver alloy is in compliance with the requirements of Section 3.

4.4.1.2 Test and inspection equipment and facilities - The manufacturer shall furnish and maintain all necessary test equipment and facilities and personnel

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for performing all quality conformance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.4.1.3 Quality conformance test procedures - Each lot of material shall be subjected to the following tests with sample size and acceptance criteria (the chemical composition test excepted) determined in accordance with MIL-STD-105, Inspection Level S-3, Acceptable Quality Level 4.0 percent.

4.4.1.3.1 Sampling procedures and acceptance criteria, dimensional, and surface finish requirements - Each lot shall be sampled and inspected for compliance with the dimensional, and surface finish requirements of this specification in accordance with MIL-STD-105, AQL 1.0 percent. Should the combined number of defective units exceed the acceptance criteria, the lot represented by these defective units shall be rejected. A single coil or strip shall be considered as a sampling unit.

4.4.1.3.2 Chemical composition - A minimum of one chemical analysis per lot shall be performed as specified in Fed. Test Method Std. No. 151 to verify that the material meets the requirements of 3.1.

4.4.1.3.3 Physical properties - Subject the samples (see 4.4.1.3) of the material to the properties tests as specified in Fed. Test Method Std. No. 151 to verify that the material is in accordance with the requirements of 3.2.

4.4.1.3.4 Bend test - Sample strips of the material shall be bent at 90 degrees to the direction of rolling, through 180 degrees around a radius equal to the thickness of the strip. Examine the material for signs of flaking or fracture. See 3.3.

4.4.1.3.5 Twist test - Subject the sample (see 4.4.1.3) of the material in wire form, having a temper of 1/4 hard or less, to 7 forward and 7 reverse 360-degree turns within a length of 3 inches. Examine for seams, slivers, or other discontinuities. See 3.4.

4.4.1.3.6 Headability test - Subject samples (4.4.1.3) of the material from each lot to a cold heading process to verify conformance to the requirement of 3.5. Examine the material for signs of flaking or fracture.

4.4.1.3.7 Microscopic examination - Samples selected in accordance with 4.4.1.3.1 shall be examined under a metallographic microscope at a suitable magnification to determine compliance with 3.6.

4.4.1.3.8 Resistivity and electrical conductivity - The test procedure for resistivity and electrical conductivity shall meet the requirements of 3.8, when tested in accord with ASTM B-193.

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4.5 Rejection -

4.5.1 Examination defects - Any sample unit having one or more defects shall be rejected. If the number of nonconforming sample units in the sample exceeds the acceptance number specified in MIL-STD-105 for that sample size, the entire lot shall be rejected, subject to the resubmittal paragraph of MIL-STD-105.

5. PREPARATION FOR DELIVERY

5.1 Length of strips - Unless otherwise specified, all strip material shall be furnished in increments of 6-foot lengths. When the contract specifies that the strips be furnished in coils, the inside diameter of the coil for material up to 0.010 inch thick inclusive shall be 4 inches minimum. For strips over 0.010 inch thick the inside diameter of the coil shall be 6 inches minimum.

5.2 Length of sheets - The length of sheet material shall be as specified in the contract or purchase order. When the contract specifies that the sheets be furnished in coils, the inside diameter of the coils shall be as specified in 5.1.

5.3 Length of wire and rod - The length of wire and rod shall be specified in the contract or purchase order.

5.4 Packaging - The material shall be packaged in Grade A paper in accordance with MIL-B-121. The overall dimensions shall be kept to their minimum without causing damage to the contents.

5.5 Marking - Each package shall be clearly marked with the manufacturer's full description of the contents, including size and quantity, in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The silver-base alloy is intended for use in the following types of applications:

(a) All types of electrical contacts such as headed rivets, springs, wire, composite forms, and in particular special assemblies as current carrying springs and reeds where the material may be required to break an electrical circuit as well as carry the current, and in sliding electrical contact applications in existing and new avionics equipment.

(b) Structural applications where mechanical strength and hardness is required with high thermal conductivity.

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6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Class and condition required (see 1.2)

(c) Dimensions required (see 3.7)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof).*

SPECIFICATION

MIL-S-22215A(AS) SILVER-COPPER-CADMIUM-NICKEL ALLOY

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED *(Printed or typed name and activity)*	DATE

DD FORM 1 NOV 64 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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